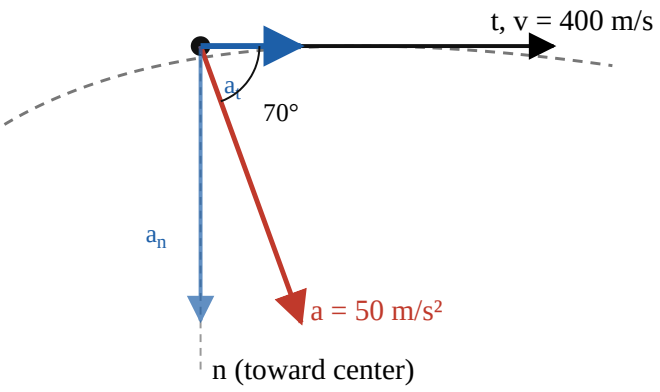
Question 1 **15 points**

CURVILINEAR MOTION – NORMAL & TANGENTIAL COMPONENTS

Given: v = 400 m/s, a = 50 m/s², with a at 70° from the direction of motion (the tangent t), pointing toward the center of curvature.
Find: the rate of increase in speed, $\dot{v} = a_t$ (5 p), and the radius of curvature ρ (10 p).



Rate of increase in speed (tangential component) – 5 p

The rate of change of speed is the tangential component of the acceleration:

$$\dot{v} = a_t = a \cdot \cos 70° = 50 \cdot \cos 70° = 50(0.3420)$$

$$\boxed{\dot{v} = a_t = 17.1 \text{ m/s}^2}$$

Radius of curvature (normal component) – 10 p

$$a_n = a \cdot \sin 70° = 50(0.9397) = 46.98 \text{ m/s}^2$$

Then use $a_n = v^2/\rho$:

$$\rho = v^2 / a_n = (400)^2 / 46.98 = 160\,000 / 46.98$$

$$\boxed{\rho = 3405 \text{ m} \approx 3.41 \text{ km}}$$

Check: $\sqrt{(a_t^2 + a_n^2)} = \sqrt{(17.10^2 + 46.98^2)} = 50.0 \text{ m/s}^2$ ✓

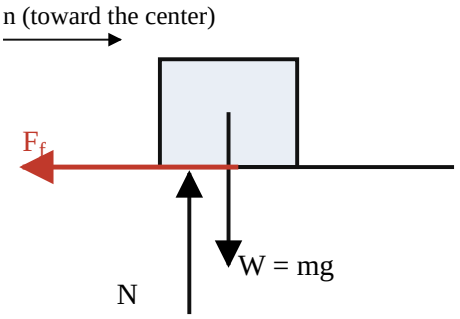
Question 2 — 20 points

EQUATION OF MOTION – NORMAL & TANGENTIAL COORDINATES

Given: m = 1 kg, constant speed v = 4 m/s, $\mu_s = 0.6$, g = 9.81 m/s²

Find: the smallest radius of curvature ρ_{min} so that the cartons do not slip.

Free-body diagram (side view)



Friction is the only horizontal force, so it must supply the centripetal force $m \cdot v^2/\rho$.

Speed is constant ⇒ $a_t = 0$

Equations of motion

$\Sigma F_b = 0$: $N - mg = 0 \Rightarrow N = mg = (1)(9.81) = 9.81$ N

$\Sigma F_t = m \cdot a_t = 0$ (constant speed)

$\Sigma F_n = m \cdot a_n$: $F_f = m \cdot v^2/\rho$

Condition for no slipping

The smallest radius is reached when friction is at its maximum (the carton is about to slip):

$F_f = \mu_s N = (0.6)(9.81) = 5.886$ N

$\mu_s \cdot m \cdot g = m \cdot v^2/\rho_{min} \Rightarrow \rho_{min} = v^2 / (\mu_s g) = (4)^2 / [(0.6)(9.81)] = 16 / 5.886$

$\rho_{min} = 2.72$ m

The mass cancels out, so the answer does not depend on the carton's mass. If $\rho < 2.72$ m, the friction needed would be more than $\mu_s N$ and the carton would slip outward.

Question 3 — 14 points

THEORY

a) Difference between kinematics and kinetics (7 p)

	Kinematics	Kinetics
Definition	Studies the **geometry of motion**: how a body moves, **without considering the forces** that cause the motion.	Studies the **relation between the forces** acting on a body and the **motion they produce**.
Quantities	Position, displacement, velocity, acceleration and time.	Force, mass, moment, momentum, work and energy, together with acceleration.
Main equations	$v = ds/dt$, $a = dv/dt$, $a\,ds = v\,dv$, $a_t = \dot{v}$, $a_n = v^2/\rho$	Newton's 2nd law $\Sigma F = m \cdot a$, work–energy, impulse–momentum
Example	Question 1: finding a_t and ρ from v and a.	Question 2: using friction (a force) to find ρ_{min}.

In short: kinematics answers *"how does it move?"* and kinetics answers *"why does it move that way?"* Kinetics uses the results of kinematics.

b) Conservation of linear momentum (7 p)

The **linear momentum** of a particle is **L = m·v** (a vector, units kg·m/s). The **principle of linear impulse and momentum** says:

$$m \cdot v_1 + \Sigma \int F\,dt = m \cdot v_2$$

If the **sum of the external impulses on a system of particles is zero**, then the total linear momentum of the system stays **constant**:

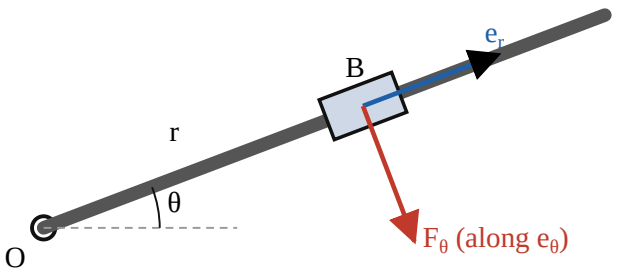
$$\Sigma\, m \cdot v_1 = \Sigma\, m \cdot v_2 \quad (\text{e.g. } m_A v_{A1} + m_B v_{B1} = m_A v_{A2} + m_B v_{B2})$$

- **Condition:** there are no external impulsive forces on the system, or they are negligible. The **internal forces** between the particles always cancel in pairs (Newton's 3rd law: equal and opposite), so they do not change the total momentum.
- Momentum is a **vector**, so it can be conserved in one direction (e.g. x) even if it is not conserved in another.
- **Applications:** collisions and impact, explosions, gun and bullet recoil, a person jumping off a boat, and coupling of railway cars.
- Note: momentum is conserved in a collision, but **kinetic energy usually is not**, because some of it is lost as heat, sound and deformation.

Question 4 **20 points**

RADIAL & TRANSVERSE (POLAR) COMPONENTS

Given: m = 300 g = 0.3 kg, r = 300 + 100 cos(0.5πt) [mm], θ = π(t² − 3t) [rad], t = 0.5 s, horizontal plane
Find: the transverse component of the force on the collar, F_θ



Step 1 – Derivatives (convert r to metres)

$r = 0.3 + 0.1\cos(0.5\pi t)$ $\quad \theta = \pi(t^2 - 3t)$

$\dot{r} = -0.05\pi \sin(0.5\pi t)$ $\quad \dot{\theta} = \pi(2t - 3)$

$\ddot{r} = -0.025\pi^2 \cos(0.5\pi t)$ $\quad \ddot{\theta} = 2\pi$

Step 2 – Values at t = 0.5 s (0.5πt = π/4, sin = cos = 0.7071)

r	$\dot{r}$	$\ddot{r}$	$\dot{\theta}$	$\ddot{\theta}$
0.3707 m	−0.1111 m/s	−0.1745 m/s²	−2π = −6.283 rad/s	2π = 6.283 rad/s²

Step 3 – Transverse acceleration

$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} = (0.3707)(6.283) + 2(-0.1111)(-6.283)$

$a_\theta = 2.329 + 1.396 = 3.725 \text{ m/s}^2$

Step 4 – Newton's 2nd law in the θ direction

$F_\theta = m \cdot a_\theta = (0.3)(3.725)$

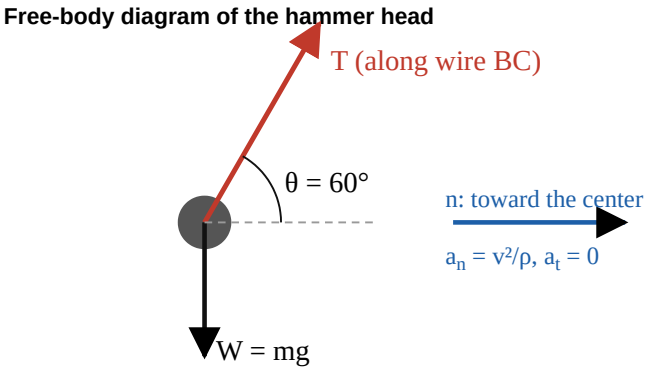
$F_\theta = 1.118$ N

(For reference, the radial part is $a_r = \ddot{r} - r\dot{\theta}^2 = -0.1745 - 0.3707(6.283)^2 = -14.81$ m/s², so $F_r = -4.44$ N. This was not asked.)

Question 5 **20 points**

UNIFORM CIRCULAR MOTION – CONICAL PENDULUM

Given: m = 7.5 kg, constant speed v = 2.5 m/s in a horizontal circle, θ = 60° (angle between wire BC and the horizontal), g = 9.81 m/s²
Find: (a) the tension T in wire BC, (b) the radius of the circle ρ



(a) Tension in wire BC

The motion is in a horizontal plane, so there is no vertical acceleration:

$$\Sigma F_y = 0: \quad T \sin\theta - mg = 0 \;\Rightarrow\; T = mg / \sin 60^\circ = (7.5)(9.81) / 0.8660$$

$T = 84.96\ \text{N} \approx 85.0\ \text{N}$

(b) Radius of the circle

The horizontal component of the tension gives the normal (centripetal) acceleration:

$$\Sigma F_n = m\,a_n: \quad T\cos\theta = m\,v^2/\rho$$

$$\rho = m\,v^2 / (T\cos\theta) = (7.5)(2.5)^2 / [(84.96)(0.5)] = 46.875 / 42.48$$

$\rho = 1.104\ \text{m}$

Check: dividing the two equations gives $\tan\theta = g\rho/v^2$, so $\rho = v^2 \cdot \tan\theta / g = (6.25)(1.732)/9.81 = 1.104$ m ✓

Question 6 **6 points**

UNIT CONVERSIONS

Acceleration: 1 mm = 10^{-3} m; 1 m = 10^{-3} km; 1 s = 1/3600 h, so 1/s² = 3600²/h² = 1.296×10^{7} /h²

a = 5 mm/s² = 5×10^{-3} m/s² = 0.005 m/s²

a = 0.005 m/s² × (1 km / 1000 m) × (3600 s / 1 h)² = 0.005 × 10^{-3} × 12 960 000

a = 5 mm/s² = 0.005 m/s² = 64.8 km/h²

Force: 1 N = 1 kg·m/s², 1 kN = 1000 N

F = 5 kN = 5 × 1000 N = 5000 N

F = 5 kN = 5000 kg·m/s²